ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · W1 · WORKING SCIENTIFICALLY

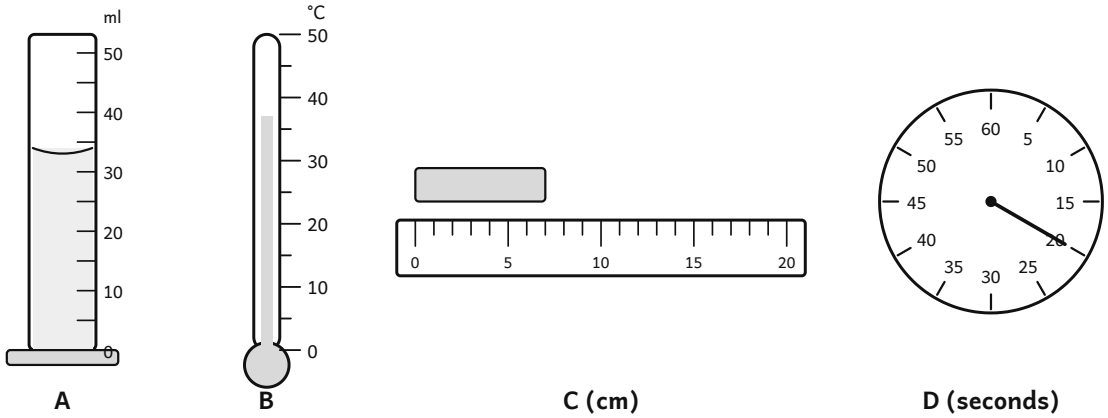
Tools of the scientist

★☆☆
Grade 5

Learning goals

- Read scales accurately.
- Match units to measurements.
- Name types of enquiry.

1 Read each instrument. Include the unit. [4]



A B C (cm) D (seconds)

Instrument	A measuring cylinder	B thermometer	C ruler	D stopwatch
Reading	34 ml	37 °C	7 cm	20 s

2 Match each measurement to its unit. [6]

Draw a line to match each pair, or write the letter.

mass of an apple	→ F	**A** decibels (dB)
force of a push	→ E	**B** lux
loudness of a horn	→ A	**C** seconds (s)
brightness of a room	→ B	**D** millilitres (ml)
volume of juice	→ D	**E** newtons (N)
time for a race	→ C	**F** grams (g)

3 **Sort the questions.** [6]

SORT THESE

Which fabric dries fastest in the sun? **Does salt make ice melt faster?**
Is red the best colour? **Which song is the most beautiful?**
Do bigger seeds grow taller plants? **Are cats nicer than dogs?**

Can be tested scientifically	Cannot be tested (opinion)
Does salt make ice melt faster?	Is red the best colour?
Which fabric dries fastest in the sun?	Are cats nicer than dogs?
Do bigger seeds grow taller plants?	Which song is the most beautiful?

4 **Write the type of enquiry: fair test, observing over time, pattern seeking, identifying and classifying or research.** [5]

Investigation	Type of enquiry
Recording the height of a sunflower every week	observing over time
Sorting leaves by their shape	identifying and classifying
Testing which paper towel soaks up most water	fair test
Reading about how vaccines were discovered	research
Do children who sleep more have faster reaction times?	pattern seeking

ANSWER KEY

InquiryMind
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THINK IT THROUGH · W1 · WORKING SCIENTIFICALLY

Fair tests, graphs and results

★★☆
Grade 5

Learning goals

- Spot unfair tests.
- Choose the right graph.
- Calculate a mean and spot anomalies.

1 **Rima wants to find out whether ice melts faster in salt water. She puts one ice cube in a cup of salt water in the sun and a bigger ice cube in a cup of tap water in the shade.** [4]

a) Why is this not a fair test?

She changed more than one thing: the size of the ice cube and the place (sun or shade) as well as the salt.

b) How should she change her plan?

Use same-sized ice cubes and the same amount of water, put both cups in the same place, and change only salt or no salt.

2 **Bar chart or line graph? Tick the best choice.** [4]

Results	Bar chart	Line graph
Height of a plant measured every day for two weeks		✓
Number of learners who like each fruit	✓	
Temperature of cooling tea every 5 minutes		✓
Water held by sand, soil and clay	✓	

3 **Tanvir timed how long a paper helicopter took to fall three times.** [3]

Try	1	2	3	Mean
Time (s)	2.1	2.3	2.2	2.2

a) Work out the mean. Show your working.

(2.1 + 2.3 + 2.2) ÷ 3 = 6.6 ÷ 3 = 2.2 s

b) Why is it better to repeat the measurement?

Single measurements can have errors; repeating and taking a mean gives more reliable results and shows anomalies.

4 **Write a prediction with a reason for this question: "Will a wet towel dry faster on a windy day or a still day?"** [2]

I think it will dry faster on a windy day because wind carries the water vapour away, so evaporation is faster.

ANSWER KEY

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INVESTIGATE · W1 · WORKING SCIENTIFICALLY

Which fabric soaks up the most water?

★★☆ Grade 5

Learning goals

- Plan a fair test with repeats.
- Identify an anomaly and calculate means.
- Present and evaluate results.

Class 5D wanted to find the best fabric for a cleaning cloth. They dipped a 10 cm × 10 cm square of each fabric in water for 30 seconds, let it drip for 10 seconds, then squeezed it into a measuring cylinder. They repeated each test three times.

1 **Complete the plan.** [4]

Question	Which fabric soaks up the most water?
Change	type of fabric
Measure	volume of water squeezed out (ml)
Keep the same	size of square, dipping time, dripping time, squeezing, same water

2 **Here are the results.** [5]

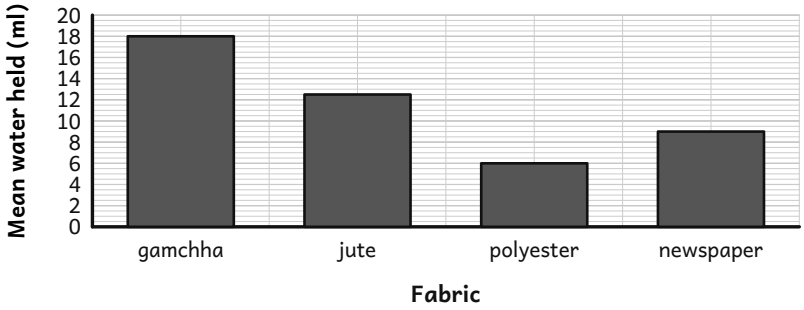
Fabric	Test 1 (ml)	Test 2 (ml)	Test 3 (ml)	Mean (ml)
cotton gamchha	18	19	17	18
jute (*pat*)	12	25	13	12.5 (without the anomaly)
polyester	6	5	7	6
newspaper	9	8	10	9

a) Circle the anomaly. Suggest a reason for it.

Jute test 2 (25 ml). Perhaps the square was bigger, left in the water longer or not allowed to drip.

b) Work out the mean for each fabric and write it in the table. Leave out the anomaly.

3 **Draw a bar chart of the means.** [3]



4 **Write a conclusion and suggest one improvement.** [3]

The cotton gamchha soaked up the most water (18 ml) and polyester the least (6 ml), so cotton is best for a cleaning cloth. Improvement: repeat the anomalous test, or test more fabrics.

ANSWER KEY

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LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · W1 · WORKING SCIENTIFICALLY

Two Boses: scientists from Bengal

★★★
Grade 5

Learning goals

- Learn how real scientists work.
- Link evidence, creativity and new instruments.
- Reflect on sharing knowledge.

Science | History | Maths | Language | Ethics

Jagadish Chandra Bose was born in **1858** in Mymensingh, in what is now Bangladesh. He studied in Kolkata, Cambridge and London, then became a professor in Kolkata. In **1895** he showed that invisible radio waves could ring a bell and set off a small explosion in another room, one of the earliest demonstrations of **wireless** signals.

Later he asked a bold question: do **plants** respond to the world as animals do? Plant growth is far too slow to see, so he invented the **crescograph**, which magnified tiny movements about **10 000 times**. His careful measurements showed that plants react to light, heat, touch and chemicals. Bose believed knowledge should be **shared** with everyone and was reluctant to make money from patents. He also wrote one of the first science fiction stories in Bengali, about a cyclone stopped by a bottle of hair oil!

Satyendra Nath Bose taught physics at **Dhaka University** from 1921. In **1924**, in Dhaka, he wrote a new way of explaining light and sent it to **Albert Einstein**, who was so impressed that he translated it into German himself. A whole family of tiny particles, **bosons**, is named after him.

Based on biographies of J. C. Bose (Bose Institute) and S. N. Bose (University of Dhaka).

1 **Put these events in order.** [2]

- 4 S. N. Bose sends his paper to Einstein (1924).
- 2 J. C. Bose demonstrates wireless signals (1895).
- 1 J. C. Bose is born in Mymensingh (1858).
- 3 S. N. Bose starts teaching at Dhaka University (1921).

2 **Maths. A plant grows 0.002 mm in one minute. The crescograph magnifies 10 000 times. How far does the pointer move in one minute?** [2]

0.002 × 10 000 = 20 mm (2 cm)

3 **Why did J. C. Bose need to invent a new instrument? What does this tell you about how science moves forward?** [2]

Plant movements were too small and slow to see or measure with existing tools. New instruments let scientists collect new evidence and answer new questions.

4 **J. C. Bose shared his discoveries freely instead of keeping them secret to make money. What are the benefits and problems of sharing?** [2]

Benefits of sharing	Problems of sharing
others can build on the ideas; science moves faster; everyone gains	the scientist may earn no money to fund more research; others may take credit

5 **Design. Invent an instrument to measure something too small, slow or fast to see. Draw it and explain what it measures.** [2]

My instrument

Any reasoned idea, e.g. a long pointer lever to magnify how much a seed swells; a sound-triggered timer for fast events.

ANSWER KEY

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REVIEW AND CHALLENGE · W1 · WORKING SCIENTIFICALLY

Show what you know

★★★ Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

M	V	Y	L	P	R	E	D	I	C	T
T	U	A	Y	L	R	D	K	O	C	G
P	W	I	R	G	W	V	W	T	G	R
A	Y	U	P	I	L	C	A	C	M	A
T	I	S	B	F	A	W	S	F	E	P
T	V	Y	A	A	L	B	D	R	A	H
E	L	A	A	I	A	G	L	H	S	O
R	V	C	S	R	U	V	B	E	U	P
N	D	T	A	B	L	E	H	U	R	R
L	H	M	O	D	E	L	W	P	E	G
O	H	P	B	R	E	P	E	A	T	A

fair **graph** **measure** **model** **pattern** **predict** **repeat** **table** **variable**

2 **Which unit measures force?** [1]

- ○ grams
- ● **newtons**
- ○ litres
- ○ degrees Celsius

3 **A result that does not fit the pattern is called...** [1]

- ○ a conclusion
- ○ a variable
- ● **an anomaly**
- ○ a prediction

4 **Draw a results table for this investigation: "How does the temperature of water change the time sugar takes to dissolve?" Include headings with units and space for three repeats.** [3]

My table

Columns: Temperature (°C) | Time 1 (s) | Time 2 (s) | Time 3 (s) | Mean time (s), with rows for each temperature.

5 **Challenge. We often draw particles in a solid as circles in neat rows. Give one way this model is useful and one limitation.** [2]

Useful: it shows that particles are close together and in a pattern. Limitation: real particles are far too small to see, are always vibrating, and are not flat circles.